Paper 02 — System Architecture

Tech Stack, Components and Data Flows

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  WHYCREMISI RESEARCH PAPERS — N.02
  Complete System Architecture
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Architecture Overview

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║                  WHYCREMISI ARCHITECTURE                  ║
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║                                                           ║
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║  │            UI LAYER  (React 18 + WebView)           │  ║
║  │  Chat · BoxChat · BotFace · Telemetry · Transport   │  ║
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║                           │  WebSocket JSON (port 8080)   ║
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║  │            BRIDGE LAYER  (C++ / JUCE)               │  ║
║  │    OscBridge · WebSocketServer · WebViewBridge      │  ║
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║  │  AiEngine    │  │  DSP Engine  │  │ AgentWorkspace  │  ║
║  │  (multi-     │  │  Analyzer    │  │ PersonalityCore │  ║
║  │  provider)   │  │  EQ/Comp/    │  │ Memory System   │  ║
║  └─────────────┬┘  │  Limiter     │  └────────┬────────┘  ║
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║  │           DAW LAYER  (JUCE AudioProcessor)          │  ║
║  │        PluginProcessor · ParameterMapper            │  ║
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║                               │  VST3 Automation / OSC    ║
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║  │          DAW HOST + THIRD-PARTY PLUGINS             │  ║
║  │  Ableton · Logic · Reaper · FabFilter · iZotope     │  ║
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Tech Stack

Layer	Technology	Rationale
UI	React 18 + Framer Motion	Reactive components, fluid animations

Styling	Tailwind CSS 4	Utility-first, zero overhead
Build	Vite 8	Fast HMR, optimised bundle
Host	JUCE 8	Industry standard for audio plugins
Bridge	WebSocket + JSON	Universal, low latency
AI	Multi-provider	Flexibility, no vendor lock-in
DSP	JUCE DSP + custom	Real-time, thread-safe
Serialisation	nlohmann/json	Header-only, fast, readable
OSC	oscpack	Standard DAW protocol

Complete Data Flow Example

Scenario: User writes "analyze the mix and tell me what's wrong with the kick"

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1. UI sends: { type:'ai.prompt', payload:{ prompt:'analyze...' } }
2. OscBridge receives, dispatches to AiEngine
3. AgentWorkspace enriches prompt:
   — BPM: 128, position: 02:34
   — Kick volume: -3dB, FFT peak at 220Hz (+2dB over ideal)
   — User memory: "prefers dry kick, narrow EQ cuts"
4. AiEngine streams response via Gemini/Claude/Ollama
5. UI renders BoxChat type 'eq' (detects "220Hz" in response)
6. User: "yes apply it"
7. ParameterMapper → FabFilter Pro-Q3: Band1 freq=220Hz gain=-3dB Q=2.1
8. AgentSoul logs: "narrow EQ cut accepted — reinforce pattern"
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→ *Continue: [Paper 03 — Third-Party Plugin Control]*